Lab 3# - Capacitive Tattoo

BME/CS 479 - Wearables and Nearables Technology Laboratory – Spring 2026 Rafael Mejia | Group 7

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Hardware Design and Development

In this project we used the FireBeetle 328P which was the brain, transmitting data via BLE to Processing UI. SparkFun MPR121 Capacitive touch sensor which has wiring to the copper tape onto the skin. The Flex sensor was attached to a glove on the finger to collect bend states. We designed both to have the peripheral board on the arm and the tattoos on the forearm. Then the Flex sensor on the finger in the glove. The second wearable was made more stylish as a vest. Some assembly challenges were fitting all 10 pads on the forearm. We also had some printing issues with the 3D model.

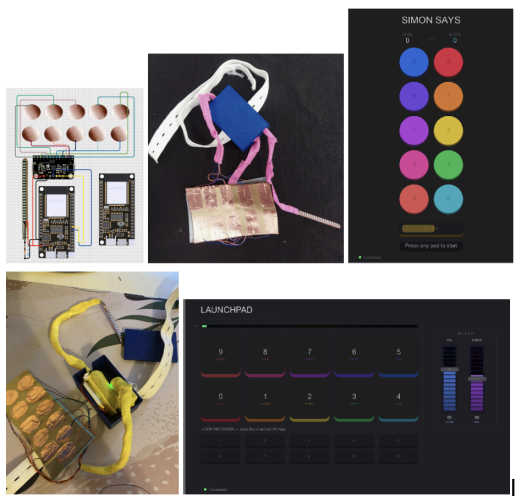

Figure: The image shows the assembled device (2 inches wide × 2 inches high) and also includes the circuit diagram and UI screenshot.

User Interface Design and Development

The user interface was designed for simplicity and to “get to the point”. For the first it was a launchpad so we made it as similar to a real one as possible and also structured the buttons on the ui the same as the tattoos. The second one was the same thing with the layout but the device was a simon says game which we added sound effects for better interaction. One thing we can improve on the code is maybe adding a feature so the user can import their own sounds to the launchpad and also add more audio features so the user can have more control over the sound.